

Lab 01 - Give Claude a Tool

30 minutes hands-on · GTM Technical Enablement
Python + the Anthropic SDK · one tool, the whole loop

The customer's real question

"Will it make things up about our data?"

- Tools are how Claude uses THEIR system of record instead of guessing.
- Step 1 shows the floor: no tool, and Claude says it can't look it up.
- The rest of the lab raises the ceiling without breaking that floor.
- By the end, you answer the question live, in code, in under a minute.

What a tool is - and isn't

- A description + a schema you hand to Claude. That's the whole thing.
- Claude never runs it. Claude asks; YOUR code executes; you send the result back.
- The description is operating instructions Claude actually follows - not docs.

request -> tool_use -> you execute -> tool_result -> answer

The loop, as the terminal prints it

step2_one_tool.py - captured from a live run

```
stop_reason: tool_use
```

```
Claude called: get_policy_status {'policy_number': 'LIC-48213'}
```

```
stop_reason: end_turn
```

```
Status: Active - Avery Chen - $500,000 - due 2026-09-15
```

Make the room say it out loud: Claude stopped to ask -> my code ran the lookup -> Claude answered from the result.

Descriptions are the product

Exercise 2b - observed, two-stage

- Full description: "policy 48213" -> Claude sends LIC-48213. Works.
- Delete the format sentence: still works - the PROPERTY example carried it.
- Delete both hints: Claude sends the bare number - and the lookup errors.
- The code never changed. The words did. Tune the description first.

Parallel calls, graceful errors

step3 - captured from a live run

```
Claude called: get_policy_status {'policy_number': 'LIC-48213'}
```

```
Claude called: get_policy_status {'policy_number': 'LIC-77102'}
```

- Two tool_use blocks in ONE turn; return both results in ONE message.
- Unknown policy -> is_error: True -> an explanation, not an invention.
- This is the demo for "what happens when your data is wrong?"

The loop by hand vs. the SDK runner

- Step 2: eleven lines, every stop_reason visible - the mental model.
- Step 4: @beta_tool + client.beta.messages.tool_runner - the loop, gone.
- The runner is BETA in the Python SDK (anthropic 1.0.0). Say so. Accuracy is the brand.
- You can only debug what you've traced once by hand.

Take it into the next customer call

- What's the one thing Claude cannot do without a tool?
- Which part of the definition do you change first when a tool misfires?
- What would you put in a tool for THEIR system of record?

Question three is the discovery conversation - a customer answering it is describing their integration, their data bound